

Mianzhi Hu

Email: mih037@ucsd.edu

Phone Number: +1 (619) 953-9445

Address: 8540 Costa Verde

LinkedIn: <https://www.linkedin.com/in/%E5%8B%89%E4%B9%8B-%E8%83%A1-0992b932b/>

Education

B.S. University of California, San Diego (Data Science Major)	07/2024-Present
The Experimental High School Attached to Beijing Normal University	09/2021-07/2024

Course List and Academic Performance at UCSD:

DSC 30 Programming and Algorithm for Data Science: **4.00**;

DSC 40a Theoretical Foundations of Data Science 1: **4.00**;

DSC 80 Practice in Data Science TBD;

MATH 180a Introduction to Probability: **4.00**;

DSC 140a (Probability Modeling and Machine Learning): Not formally enrolled;

DSC 291 (Cognitive Science and NLP): Not formally enrolled;

DSC 291 (Machine Learning with Few Labels): Not formally enrolled;

Skills

Python and surrounding skills: Python, Pytorch, Numpy, Pandas, Pillow, COCO Tools;

Web technology: Comprehension of IIS server tech; SQL for database; PHP for text loading, user registration and payment services; Javascript and NodeJS;

Programming with Java: Searching, hash, DP, graph theory, data structures, prefix algorithms, etc.;

AI theories and technology: Image augmentation, classification, segmentation and generating tech; large-scale image data processing and augmentation design; transformers and multimodal AI;

Capacity to work on server clusters: basic knowledge about Virtual Machine and Linux; Nautilus and Kubernetes and Jupiter; familiarity with Colab;

Maths, Statistics and Machine Learning: linear algebra, calculus and statistics; maths-required algorithms for Machine learning.

Experiences in DATA SCIENCE

Research on AI Video Adjusting Model for Personalized Advertisements

Researcher

03/2025-Present

- With multi-modal representation system focusing on modal merging on language-language and language-image sequences, designed data processing methods and tested it with CCA landmarks
- Used BERT&Stanza to build an AI semantic comprehension model for commands, processed training and testing datasets;
- Collected 10000+ video data from online with Python and processed some with Pytorch, applied augmentation algorithms such as Sobel, Gaussian, rotation, blurring, etc. on large datasets such as Youtube8M
- Now investigating and building models for integrated multi-modal representation, semantic segmentation and physical constraining technique to ensure stability; fine-tuning the model with personalized data (400+ sentence commands as datasets).

Research on AI Gaze Tracking at Prof. De Sa's Cognitive Science Lab at UCSD

Researcher

06/2025-Present

- With Convolution LSTM and Hourglass network, designed a model generating heat-maps and 3D landmarks for eyes images;
- Designed a CVAE network for calibration and personalized data stability with Semi-supervised Semantic Alignment mechanics, as well as group sparsity of features; trying to add personalized fine tuning tools;
- Generated 8000+ eye data with UnityEyes and collected real word datasets; designed data polluting functions with Numpy for model robustness, including head pose changing, swirling, Gaussian filters, upsizing, etc; fleshed out the datasets to over 20000 pictures
- Learning to add computer graphics to refine model stability on low-quality, multi-target pictures and designing pre-calibration model by computer graphic knowledge such as Retinex and SH.

Research on Image Generating & Filtering Model for Novel at Tsinghua University

Researcher

05/2023-09/2023

- Learned text processing technology such as snowNLP and AI image generation algorithms such as BERT; collected and augmented 600+ texts (with grammar trees selecting key words) and 1300+ pictures (with Numpy and self-made style transplanting model with CNN) for model training;
- Built a novel illustration generation model with BERT & VGG; filtering out pornographic or violent results with Resnet and related models;
- Wrote a 11-pg paper; Presented at 20233rd Intl. Workshop on Frontiers of Graphics & Image Processing;
- Now Handing on refining the generation-filtering system with small batched, data-efficient semi-supervised learning and apply it to proper presentations generation, bias / false image / irrational prompts filtering;

Website Including Registration and Publication System (www.luomiart.com)

Developer

08/2024-Present

- Designed datasets with SQL, accommodating more than 500 articles and 100 users; updated the datasets for few times while learning snowflake structure; built datasets with relationship between publication, registration payment and location data;
- With PHP, wrote code to load dynamic (small animate resources) and static resources (pictures and videos); re-organized some back end pieces such as asynchronous presentation of comments;
- Learning ETL designing system; trying to rebuild the JAVA based data management system on Kafka (for event data capture), Spark (data processing and loading to optimized BigQuery), and Airflow (for task organizing).